Personal Learning Journey – Withanan W.Y.P.

Student	Student ID	Group	Member	Role
Withanan W.Y.P.	IT24102008	2026-AI-33	Member 2	Preprocessing & Feature Engineering

My assigned role is to turn the understood dataset into consistent inputs for regression models. The main lesson from this stage is that preprocessing decisions must follow variable meaning and must be fitted at the correct point in validation.

Contribution and evidence

The preprocessing notebook retains all 395 students and separates target G3 from the inputs. The main scenario excludes G1/G2, leaving 30 raw features. The reproducible split assigns 316 students to training and 79 to testing. The implementation distinguishes 13 binary, four unordered nominal and ten ordinal features, plus age, failures and absences. Binary mappings are explicit, nominal variables use one-hot encoding, and ordered codes are retained numerically.

A decision that shaped the project

Right-skewed absences motivated comparison of the original values with log1p(absences). The transform preserves zeros and compresses the upper tail, but a more symmetric distribution does not itself prove improved prediction. Both representations are therefore evaluated using training cross-validation. Large absence values remain because they are not confirmed errors. No imputation is needed for the current complete dataset.

Learning and critical judgement

Standardization is appropriate for the numerical and ordered inputs of Linear Regression and Ridge; the tree models do not require it. Binary and one-hot features remain separate. The important validation lesson is to attach preprocessing to the estimator so each fold learns its own encoding and scaling. Fitting learned transformations on all data first would let held-out information influence validation. Treating ordinal codes as numeric also introduces a spacing assumption that deserves acknowledgement.

Team handoff and next steps

The shared make_preprocessor factory gives Member 3 a repeatable pipeline, including optional scaling, log absences and later-stage G1/G2 inclusion. Alignment, finite-value and column-consistency checks support this handoff. The 43-column demonstration uses full one-hot encoding; modelling uses reference-category encoding, producing 39 columns without prior grades and 41 with them. My next priorities would be confirming absence timing and the ambiguous source description of failures before operational interpretation.

Personal reflection

The check that taught me most was the column count. The demonstration produced 43 columns but the model pipelines produced 39, which first looked like a lost feature. Counting by group resolved it: 13 binary, 17 one-hot levels across the four nominal features, ten ordinal and three other numeric inputs give 43, and drop=first removes one reference level from each nominal feature, giving 39; adding G1 and G2 gives 41. I now explain an output shape from the encoding rules instead of assuming an error.

Evidence and AI use

Evidence: notebooks/02_preprocessing_feature_engineering_ipynb.ipynb; src/preprocessing.py. I used an AI assistant to help debug the preprocessing pipeline code, mainly ColumnTransformer and FunctionTransformer errors. I kept a suggested fix only after rerunning the notebook's alignment, finite-value and column-consistency checks. AI also helped draft this report; its figures were checked against the notebook outputs.